

1 V = libovolný komplexní v. pr. , $\langle \cdot, \cdot \rangle$ sk. s. , pro $\forall \vec{u}, \vec{v} \in V$:

Dokažte : 1) $\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$, $\operatorname{Re}(x)$ - reálná část komp. čísla x

2) $\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2)$, $\operatorname{Im}(x)$ - imag. část komp. čísla x

$$1) \frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2) \xrightarrow{\text{def. normy}} \frac{1}{4} (\langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle - \langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle) \xrightarrow{\text{linearita sk. s.}}$$

$$\frac{1}{4} (\langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle - (\langle \vec{u}, \vec{u} \rangle - \langle \vec{u}, \vec{v} \rangle - \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle)) \xrightarrow{\text{"skotosymetrie" + def. normy}}$$

$$\frac{1}{4} (\|\vec{u}\|^2 + \langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle} + \|\vec{v}\|^2 - (\|\vec{u}\|^2 - \langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle} + \|\vec{v}\|^2)) \xrightarrow{\text{úpravy}}$$

pro libovolná dvě komplexní čísla $z, \bar{z} \in \mathbb{C}$ platí

$$\left[\begin{array}{l} \text{pokud } z = a + bi \\ \bar{z} = a - bi \end{array} , \text{ pro } a, b \in \mathbb{R} \text{ pak } z + \bar{z} = a + bi + a - bi = 2a \Rightarrow 2\operatorname{Re}(z) \right]$$

$$\frac{1}{4} (\|\vec{u}\|^2 - \|\vec{u}\|^2 + \|\vec{v}\|^2 - \|\vec{v}\|^2 + 2\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) + 2\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)) = \frac{1}{4} (4\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)) = \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle)$$

$$2) \frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2) \xrightarrow{\text{def. normy}} \frac{1}{4} (\langle \vec{u} - i\vec{v}, \vec{u} - i\vec{v} \rangle - \langle \vec{u} + i\vec{v}, \vec{u} + i\vec{v} \rangle)$$

vytáhneme z 1. složky "skotosymetrie" + linearita

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$$- (\langle \vec{u}, \vec{u} \rangle + i\langle \vec{u}, \vec{v} \rangle + i\overline{\langle \vec{u}, \vec{v} \rangle} + \langle \vec{v}, \vec{v} \rangle) = \frac{1}{4} (-i\langle \vec{u}, \vec{v} \rangle + i\overline{\langle \vec{u}, \vec{v} \rangle} - (i\langle \vec{u}, \vec{v} \rangle - i\overline{\langle \vec{u}, \vec{v} \rangle}))$$

pro libovolná dvě komplexní čísla $z, \bar{z} \in \mathbb{C}$ platí

$$\left[\begin{array}{l} \text{pokud } z = a + bi \\ \bar{z} = a - bi \end{array} , \text{ pro } a, b \in \mathbb{R} \text{ pak } -iz + i\bar{z} = i(-z + \bar{z}) = i(-a - bi + a - bi) = 2b$$

$$iz - i\bar{z} = i(z - \bar{z}) = i(a + bi - a + bi) = -2b$$

$$= \frac{1}{4} (2\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) - (-2\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle))) = \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle)$$

Využili jsme vlastnosti komplexních čísel a komplexně sdružených v [...].

